

EDUCATION PATH AI



AI-BASED MAJOR RECOMMENDATION SYSTEM

OBJECTIVES

- Assist students in selecting majors that match their interests and abilities.
- Provide objective recommendations using Artificial Intelligence.
- Reduce the risk of inappropriate major selection.
- Support school counselors in the career guidance process.

KEY FEATURES

- Student profile analysis based on the RIASEC model.
- Personalized major recommendation system.
- Transparent explanation for each recommendation.
- Learning roadmap aligned with career goals.

ETHICAL ANALYSIS

- Privacy: Student data is processed only with user consent and is not permanently stored without authorization.
- Fairness: The system provides recommendations objectively without discrimination based on gender, socioeconomic background, or other personal characteristics.
- Transparency: Every recommendation is accompanied by a clear explanation of the factors influencing the AI's decision.
- Human Oversight: Students are encouraged to consult with school counselors before making final educational decisions.

ABSTRACT

Choosing a university major is a crucial decision that significantly influences students' academic and career paths. However, many students struggle to make informed choices due to limited self-awareness, insufficient career information, and external influences such as trends or parental expectations. Education Path AI is proposed as an Artificial Intelligence-based decision support system that analyzes academic performance, interests, and RIASEC personality profiles to generate personalized major recommendations. The system aims to provide transparent and objective guidance while supporting, rather than replacing, school counselors. Ethical principles, including data privacy, fairness, transparency, and human oversight, are integrated into the system design to ensure responsible AI implementation.

SYSTEM ARCHITECTURE

INPUT

- Academic Scores
- Interests & Achievements
- RIASEC Assessment

PRE PROCESSING

- Feature Extraction
- RIASEC Score Calculation
- Feature Normalization

AI RECOMMENDATION ENGINE

- K-Nearest Neighbors (KNN), K=5, KD-Tree
- Similarity Matching from The Closest 5 Metric Profiles
- Major Ranking

OUTPUT

Top 3 Major Recommendation

GITHUB



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